AMBASSADOR *of* PAKISTAN'S ACHIEVEMENT AWARD 2025

Dr. Ifra Ferheen

for Outstanding Achievement in the field of

SCIENCE

His Excellency Ali Javed

Rome , 16 September 2025

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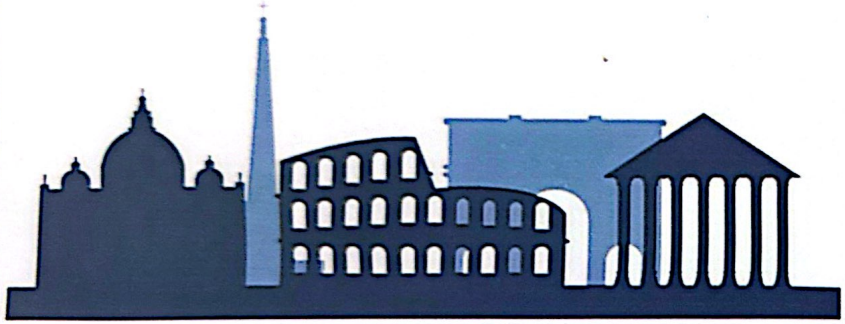
Rome, 11th October 2023

https://italianbranch.setac.org/events/



2nd Workshop

SETAC®

hosted by

Consiglio Nazionale delle Ricerche

Italian Language Branch

Best Young Italian Scientist Award

1° Classificato

Claudia Vay

IFRA FERHEEN

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The Federation of European Microbiological Societies

CONGRATULATES

Ifra Ferheen

Recipient of the *FEMS Microbes*
Poster Prize

ON THE OCCASION OF

FEMS2023, 10th Congress of European Microbiologists,
Hamburg, Germany, 9-13 July 2023

FEMS

Dr. Jana Jass
Editor-in-Chief
FEMS Microbes

Dr. Kimberly Kline
Editor-in-Chief
FEMS Microbes

Dr. Kathleen Scott
Editor-in-Chief
FEMS Microbes

MICROBES
FEMS

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iSAS
nternational
nterdisciplinary
ntersectoral
School of Advanced Studies

HR EXCELLENCE IN RESEARCH

1th PhD WORKSHOP
LIFE AND HEALTH SCIENCES

BEST PRESENTATION AWARD 2022

Camerino, November 25th 2022

has been attributed to

IFRA FERHEEN

for the presentation entitled

Plastics as Durable Islands for Harboring Multidrug-Resistant Bacteria in Aquatic Ecosystems

Prof. Anna Maria Eleuteri
PhD course Coordinator

IFRA FERHEEN

Nationality: Pakistani
(+39) 3491133439
Date of birth: 15/06/1995
Gender: Female

Email address: ifra.ferheen@iss.it

Address: Viale Regina Elena, 299, 00161, Rome, Italy

EDUCATION AND TRAINING

Doctor of Philosophy (Life and Health Sciences: Molecular Biology and Cellular Biotechnology)

Final Remarks: Excellent
University of Camerino, Camerino, Italy [16/10/2020 – 11/07/2024]
Other Affiliation: National Institute of Health, Rome, Italy

Thesis: “Analysis of the role of plastics in spreading potential pathogenic microorganisms and their antibiotic resistance genes in aquatic ecosystems, using microbiological and molecular methods”

Masters of Science (Microbiology and Immunology)

COMSATS University Islamabad, Islamabad, Pakistan [15/10/2017 – 15/07/2019]

Final grade: CGPA: 3.72/4.00

Thesis: "Cryophilic bacteria as a source of biotechnologically important proteins"

Bachelors of Science (Biosciences)

University of Wah, Wah Cantt, Pakistan [15/09/2013 – 15/06/2017]

Final grade: CGPA: 3.24/4.00

Research Report: “Resistivity pattern of *Staphylococcus aureus* against betalactam antibiotics and use of silver nanoparticles as nanotherapeutics"

WORK EXPERIENCE

Researcher Level III

National Institute of Health, Rome, Italy [18/06/2024 – Current]

Project: Protecting human health by understanding how climate change can impact pathogens and antimicrobial drug-resistant bacteria in coastal waters.

Guest Researcher

BC3 Basque Center for Climatic Change, Leioa, Spain [05/05/2023 – 31/08/2023]

Project: Effect of Land use decisions on the abundance of multi-drug-resistant bacteria and Antibiotic-Resistant in Bracciano lake, a case study

Doctorate Researcher

National Institute of Health, Rome, Italy [10/08/2020 – 15/01/2024]

Project: Field-based analysis of the spread of AMR genes mediated by Artificial Plastic Substrates in aquatic ecosystem

Research Internee

Allama Iqbal Open University, Islamabad, Pakistan [10/06/2019 –10/10/2019]

Expertise acquired: Basic microbiological and molecular biology techniques

Guest Researcher

University of Camerino, Camerino, Italy [10/10/2018 – 20/12/2018]

Expertise acquired: Antifreeze protein, Exopolysaccharide isolation and characterization for investigating its role in cryoprotectant activity for probiotic bacteria

Research Internee

COMSATS University Islamabad, Islamabad, Pakistan [01/06/2018 – 30/09/2018]

Project: Development of molecular diagnostic method for detection of *Coxiella burnetii*

RESEARCH SKILLS

Molecular Techniques: DNA Extraction (from environmental and tissue samples), Polymerase Chain Reaction, Real-time PCR for genotypic profiling of bacterial isolates, Agarose Gel Electrophoresis, SDS-PAGE, Plasmid construction, Transformation Assays, Immunoaffinity Precipitation techniques, ELISA operations, and Exopolysaccharide extraction.

Microbiological Techniques: Screening and isolation of bacteria, aseptic techniques, culturing techniques, enumerating bacteria, and identifying different classes of microorganisms and antimicrobial susceptibility assays (Kirby baeur disc diffusion assay, Minimum Inhibitory concentration).

Bioinformatics tools: Clustal W, T coffee, Primer 3, BLAST, MEGA11.

Data Analysis tools: ORIGIN Pro 2019, SPSS, MS-Excel, Geographical Information System software (ARC-GIS tool), Canva Pro, and Biorender.

PUBLICATIONS (Grey Highlighted publications are from PhD. Thesis)

1: **Ferheen, I.,** Cimarelli, L., Marcheggiani, S., Klümper, U., & Spurio, R. (2025). Plastic-mediated transformation: A new route to navigate plasmid-borne antibiotic resistance genes. Science of The Total Environment, 976, 179125.

2: **Ferheen, I.,** Spurio, R., & Marchggiani, S. (2025). Protocol for sampling multidrug-resistant bacteria using artificial plastic substrates in aquatic ecosystems. STAR protocols, 6(2), 103822.

3: **Ferheen, I.,** Spurio, R., & Marcheggiani, S. (2024). Emerging Issues on Antibiotic-Resistant Bacteria Colonizing Plastic Waste in Aquatic Ecosystems. Antibiotics, 13(4), 339.

4: **Ferheen, I.,** Spurio, R., & Marcheggiani, S. (2024). Vehicle transmission of antibiotic-resistant pathogens mediated by plastic debris in aquatic ecosystems. iScience, DOI:https://doi.org/10.1016/j.isci.2024.110026.

5: Liaqat, S., Ahmed, Z., Umer, M. U., Ali, Q., Mustafa, M. F., **Ferheen, I.,** & Waseem, M. (2024). Sheep bone valorization: Enhancing gastronomic sustainability through hydroxyapatite-enriched potato wedges. International Journal of Gastronomy and Food Science, 35, 100841.

6: Amin, A., Noureen, R., Iftikhar, A., Hussain, A., Alonazi, W.B., Raza, H.M.Z., **Ferheen, I.,** & Ibrahim, M. (2024). Uropathogenic bacteria and deductive genomics towards antimicrobial resistance, virulence, and potential drug targets. International Microbiology, 27 (1), 325-335.

7: **Ferheen, I.,** Spurio, R., Mancini, L., & Marcheggiani, S. (2023). Detection of *Morganella morganii* bound to a plastic substrate in surface water. Journal of Global Antimicrobial Resistance, 104-107.

8: **Ferheen, I.,** Ahmed, Z., Alonazi, W. B., Pessina, A., Ibrahim, M., Pucciarelli, S., & Bokhari, H. (2023). Diverse Repertoire and Relationship of Exopolysaccharide Genes in Cold-Adapted *Acinetobacter sp.* CUI-P1 Revealed by Comparative Genome Analysis. Microorganisms, 11(4), 885.

9: Afreen, A., Ahmed, Z., Khalid, N., **Ferheen, I.,** & Ahmed, I. (2023). Optimization and cholesterol-lowering activity of exopolysaccharide from *Lactiplantibacillus paraplantarum* NCCP 962. Applied Microbiology and Biotechnology, 1- 16.

10: Rashid, F., Ahmed, Z., **Ferheen, I.,** Mehmood, T., Liaqat, S., Ghoneim, M. M., & Rahman, A. (2024). Effect of fenugreek and flaxseed polysaccharide-based edible coatings on the quality attributes and shelf life of apple fruit during storage. Food Science & Nutrition, 12(3), 2093-2103.

11: Ahmed, Z., Mehmood, T., **Ferheen, I.,** Noori, A. W., Almansouri, M., & Waseem, M. (2023). Optimization of exopolysaccharide produced by L. Kefiranofaciens ZW3 using response surface methodology. *International Journal of Food Properties*, *26*(1), 2285-2293.

12: Fatima, A., Ibrahim, M., Naseer, A., Pervez, A., Asad, M., Shah, A., Hasan, F., Alonazi, WB., **Ferheen, I.,** Khan, S. (2023). Identification, genome sequencing, and characterizations of *Helicobacter pylori* sourced from Pakistan. Microorganisms.; 11(11):2658. https://doi.org/10.3390/microorganisms11112658.

13: Syed, T., Batool, U., Aslam, M., Noreen, Z., **Ferheen, I.,** Gondal, A., & Bokhari, H. (2019). Bioremediation and decontamination potential of flagellate *Poteriospumella* sp. BioremediationJournal, 1-12.

CONFERENCES AND SEMINARS

1: On-site presentation in 34th SETAC Annual Meeting, Seville

Ferheen, I., Hernandez, O., Chiabai, A., Neumann, M., Spurio, R., Marcheggiani., S., (7th May 2024) Effect of Land Use decisions on the abundance of Multi drug resistant bacteria and Antibiotic-Resistant in Bracciano Lake**,** SETAC Europe 34th Annual Meeting, Seville, Spain

2: Participation in 42nd Summer school, Bilbao

Ferheen, I., VITALISE 2023 Summer School on Living Labs. June 08, 2023 to June 09, 2023, within the 42nd Summer Courses organized by the University of the Basque Country, Bilbao, Spain

3: On-site presentation in 2nd workshop SETAC, Rome

Ferheen, I., Spurio, R., Marcheggiani, S. Detection of Extended-Spectrum Beta-lactamase producing Klebsiella spp., adhering plastic debris in Bracciano Lake, (2023), SETAC Italian Language Branch, 2nd workshop, Rome, Italy

4: On-site poster presentation in FEMS 10th Congress of European Microbiologists, Hamburg

Ferheen, I., Cimarelli L., Spurio, R., A Novel Framework to Monitor the Role of Plastic Debris in Horizontal Gene Transfer (2023), FEMS Conference on Microbiology, Hamburg, Germany.

5: On-site presentation in 1st PhD workshop Life and Health Sciences, Camerino

Ferheen, I., Spurio, R., Marcheggiani, S., (2022) Plastics as Durable Islands for Harboring Multidrug-Resistant Bacteria in Aquatic Ecosystem, Ist PhD Workshop, University of Camerino, Camerino, Italy.

6: Online presentation in 33rd Annual Meeting SETAC, Dublin

Ferheen, I., Spurio, R., Marcheggiani, S. Detection of Multidrug-resistant *Staphylococcus spp.* bound to plastic substrates in surface

water, (2022), SETAC Europe 33rd Annual Meeting, Dublin, Ireland.

7: Online presentation for Biómicas Project, Lima

Ferheen, I., Plastics as durable islands for harboring multidrug-resistant bacteria in aquatic ecosystems (2022), Biómicas Project, Peruvian University of Applied Sciences, Lima, Perú.

8: On-site presentation in FEMS Conference on Microbiology, Belgrade

Ferheen, I., Spurio, R., Marcheggiani, S. Plastics in aquatic ecosystems: a new route for trafficking multidrug-resistant bacteria? (2021), FEMS Conference on Microbiology, Belgrade, Serbia.

9: Online presentation for young researchers bilateral China Jiliang University

Ferheen, I., Isolation and Identification of Bacterial Pathogens from Plastic Bottles Floating in Freshwater (2021) Bilateral Meeting China Jiliang University.

10: Online presentation, bilateral doctoral meeting Jilin Agricultural University (Changchun, China)

Ferheen, I., Comparative genomic analysis and characterization of cold-adapted *Acinetobacter sp.* for its exopolysaccharide production and cryoprotective activity (2021) Bilateral Meeting China Jiliang University.

AWARDS AND HONOURS (Grey highlighted won during PhD thesis)

1: Recipient of Meeting Attendance Grant, **May 2024**, Society of Environmental Toxicology and Chemistry, for 34th Annual Meeting SETAC2024, Seville, Spain.

2: Recipient of Best Young Italian Scientist Award, October 2023, Second Workshop, SETAC Italian Language Branch, presentation on "Detection of ESBL-Producing *Klebsiella* spp. adhering plastic debris in Bracciano Lake".

3: Recipient of FEMS Microbes Best Poster Prize, July 2023, FEMS2023, 10th Congress of European Microbiologist,

Hamburg, Germany. Presented poster on "Role of Plastic debris in triggering horizontal gene transfer in aquatic ecosystems".

4: Recipient of Meeting Attendance Grant, **July 2023**, Federation of European Microbiological Societies, for FEMS2023, Hamburg, Germany.

5: Recipient of Best Presentation Award, November 2022, Ist PhD Workshop Life and Health Sciences, University of Camerino, Italy, presentation on "Plastics as Durable Islands for Harboring Multidrug-Resistant Bacteria in Aquatic Ecosystem".

6: Marie Sklodowska-Curie Research and Innovation Staff Exchange (RISE), European Horizon 2020 program (2018), Secondment for Master Thesis scholarship funded by METABLE (number 645693), University of Camerino, Italy.

Projects

1: European Union Project: **BlueAdapt:** Protecting human health by understanding how climate change can impact pathogens and antimicrobial drug resistant bacteria in coastal waters.

2: European Union Project: **EU-JAMRAI 2:** joint action antimicrobial resistance and Health0care associated infections

3: European Union Project: **SURE-AMR**: Strengthening Ukraine's research excellence in the field of antimicrobial resistance under a One Health Approach

4: European Union Project: **TULIP:** Community-based engagement and interventions to stem the tide of antimicrobial resistance spread in the aquatic environments catalysed by climate change and plastic pollution interactions

REFERENCES

Reference:
Dr. Stefania Marcheggiani
Researcher
Higher Institute of Health,
Rome Italy
stefania.marcheggiani@iss.it
External Supervisor PhD. Thesis